

Chapter 14 Study Guide

Key Concepts

14.1 Properties of Gases

- Gases are easily compressed because of the space between particles in a gas.
- The amount of gas (n), volume (V), and temperature (T) are factors that affect gas pressure (P).

14.2 The Gas Laws

- As the pressure of a gas increases, the volume decreases, if the temperature is constant.
- As the temperature of an enclosed gas increases, the volume increases, if the pressure is constant.
- As the temperature of an enclosed gas increases, the pressure increases, if the volume is constant.
- The combined gas law allows you to do calculations for situations in which only the amount of gas is constant.

14.3 Ideal Gases

- To calculate the number of moles of a contained gas requires an expression that contains the variable n .
- Real gases differ most from an ideal gas at low temperatures and high pressures.

14.4 Gases: Mixtures and Movements

- In a mixture of gases, the total pressure is the sum of the partial pressures of the gases.
- Gases of lower molar mass diffuse and effuse faster than gases of higher molar mass.

Vocabulary

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| ● Boyle's law (p. 418) | ● effusion (p. 435) |
| ● Charles's law (p. 420) | ● Gay-Lussac's law (p. 422) |
| ● combined gas law (p. 424) | ● Graham's law of effusion (p. 435) |
| ● compressibility (p. 413) | ● ideal gas constant (p. 426) |
| ● Dalton's law of partial pressures (p. 432) | ● ideal gas law (p. 426) |
| ● diffusion (p. 435) | ● partial pressure (p. 432) |

Key Equations

Boyle's law: $P_1 \times V_1 = P_2 \times V_2$

Charles's law: $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

Gay-Lussac's law: $\frac{P_1}{T_1} = \frac{P_2}{T_2}$

Combined gas law: $\frac{P_1 \times V_1}{T_1} = \frac{P_2 \times V_2}{T_2}$

Ideal gas law: $P \times V = n \times R \times T$ or $PV = nRT$

Dalton's law: $P_{\text{total}} = P_1 + P_2 + P_3 + \dots$

Graham's law: $\frac{\text{Rate}_A}{\text{Rate}_B} = \sqrt{\frac{\text{molar mass}_B}{\text{molar mass}_A}}$